

Geometrical considerations for design of surface mines

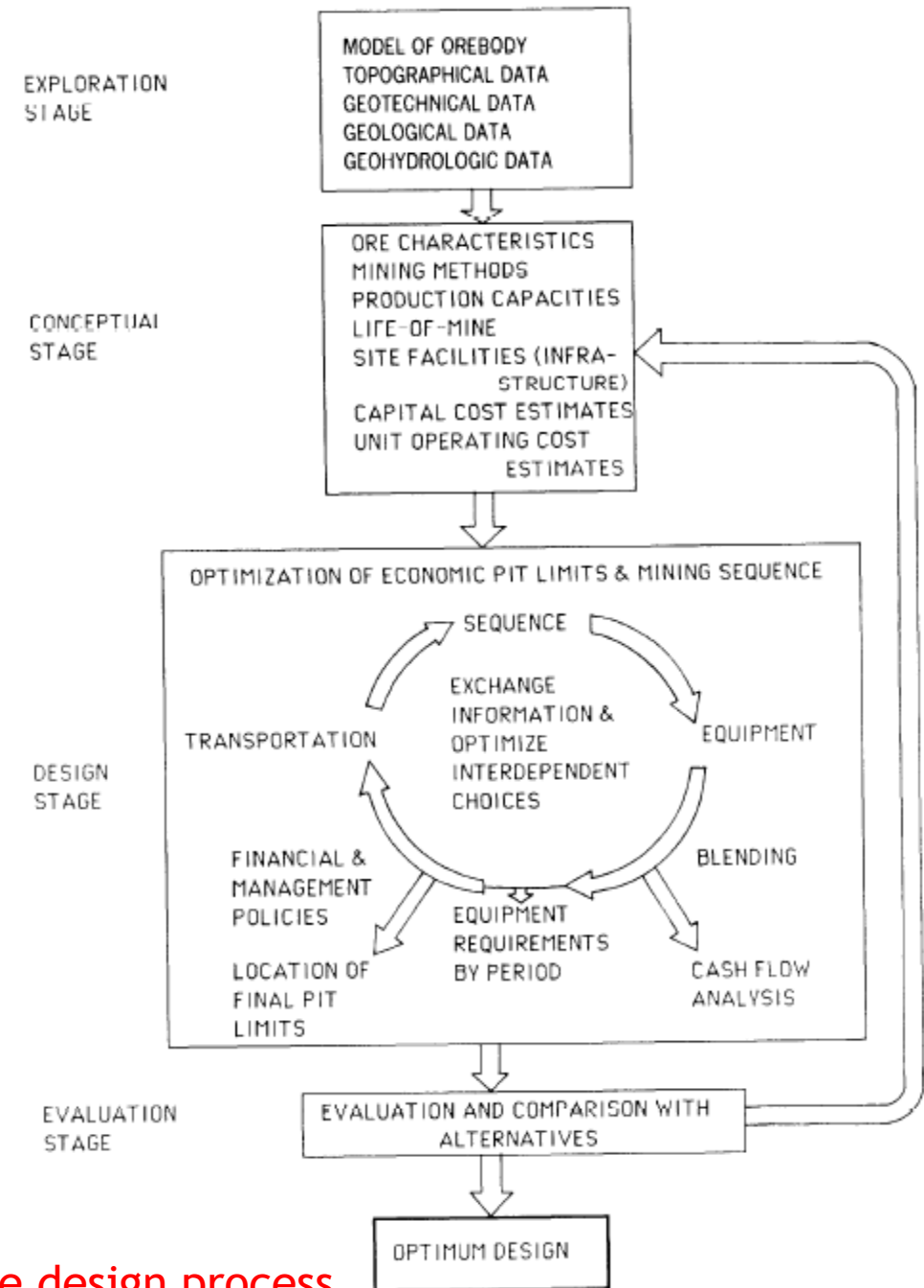
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Geometrical considerations for design of surface mines

- ▶ Pit geometry,
- ▶ Pit layouts
- ▶ Mine access,
- ▶ Pit expansion and
- ▶ Push back operation.

Pit geometry-Introduction

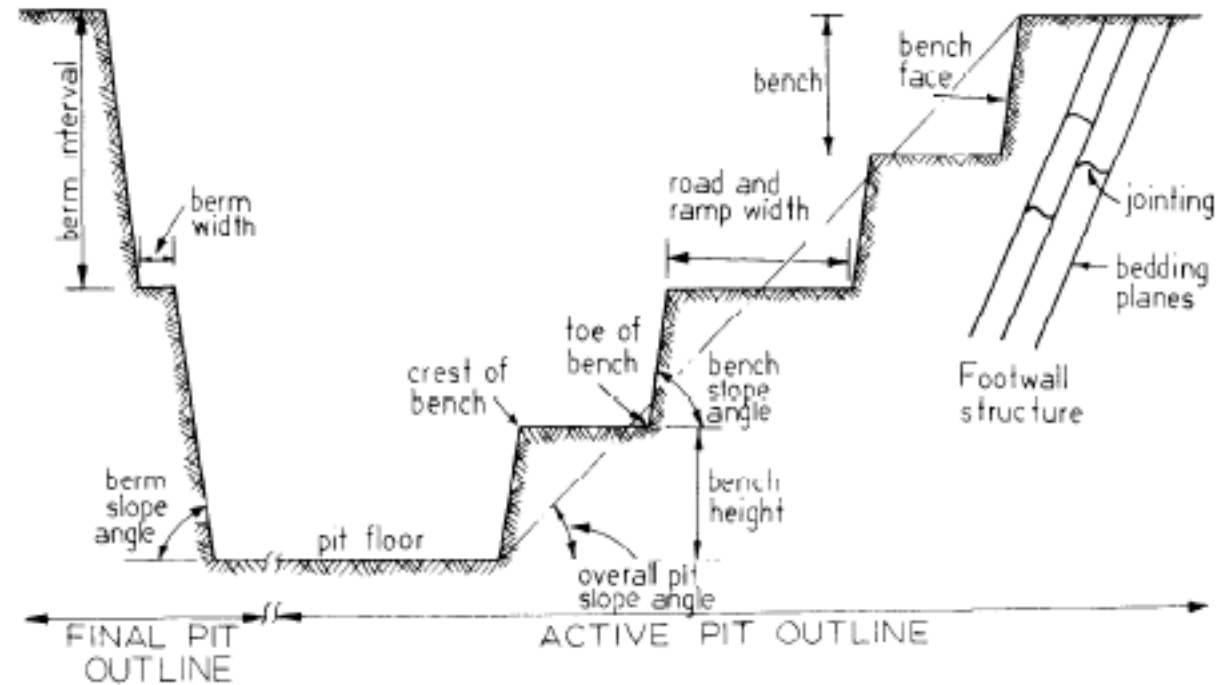
- ▶ An open pit mine is an excavation or cut made at the surface of the ground for the purpose of extracting ore and which is open to the surface for the duration of the mine's life.
- ▶ It is generally conceded that surface mining is more advantageous than underground mining in terms of recovery, grade control, economy, flexibility of operation, safety, and the working environment.
- ▶ Open pit design is conducted in several stages. They consist, technically, of devising a scheme or set of alternative schemes, followed by an evaluation and selection of the optimum scheme, as illustrated in Fig.
- ▶ The most economic final pit design often depends on factors that are largely outside the mining engineer's control, such as the geometric outline of the ore body, the distribution of ore within the ore body, topography, maximum allowable slope angles, etc. The economics of the mining program, however, depends upon the choice of mining ratio, production rates, and equipment, all of which are determined by the mining engineer.



Diagrammatic presentation of the design process

Open Pit Terminology

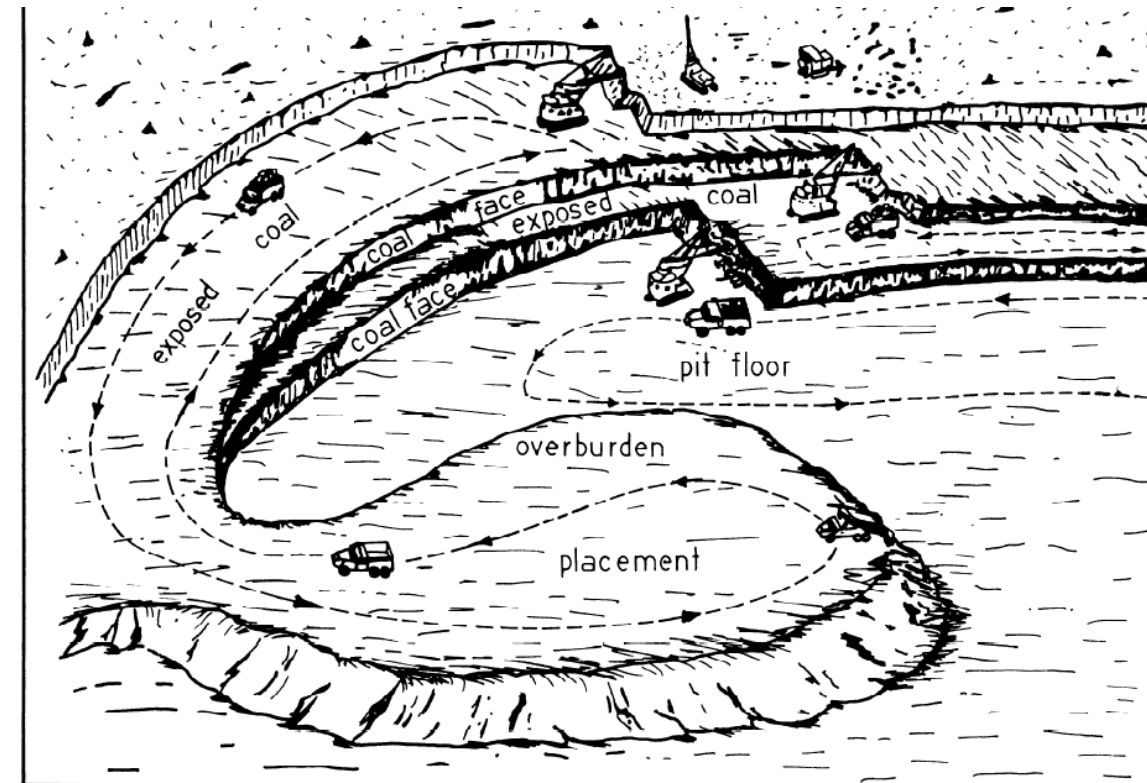
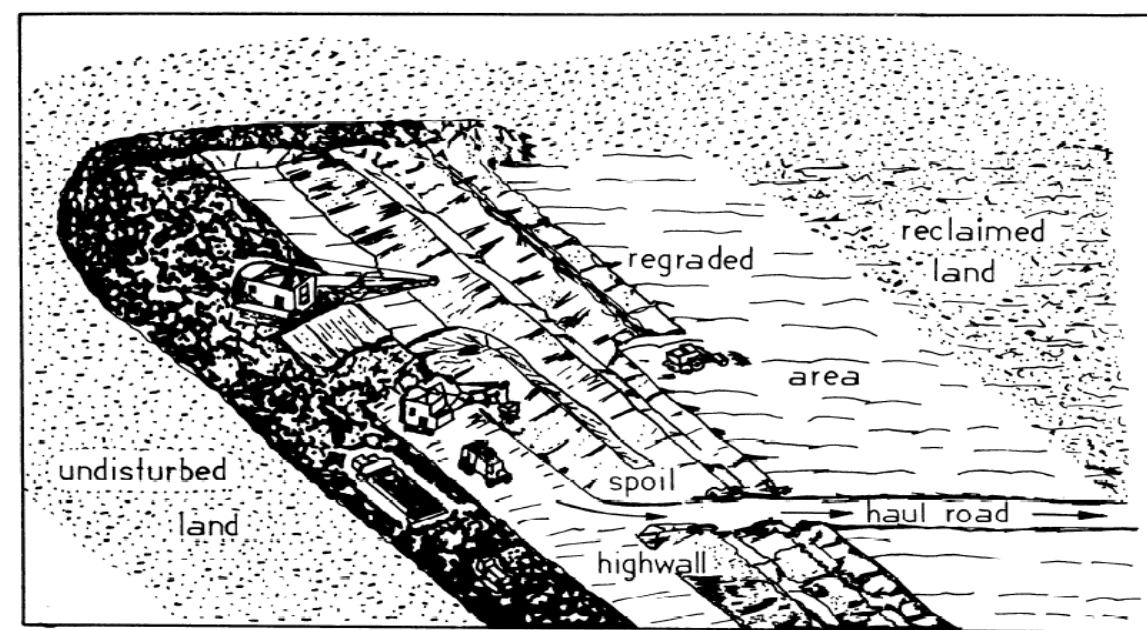
- ▶ A bench may be defined as a ledge that forms a single level of operation above which mineral or waste materials are mined back to a bench face.
- ▶ The bench height is the vertical distance between the highest point of the bench, or the bench crest, and the toe of the bench
- ▶ The bench slope is the angle, measured in degrees, between the horizontal and an imaginary line joining the bench toe and crest.
- ▶ Pit limits are the vertical and lateral extent to which the open pit mining may be economically conducted.
- ▶ berm is a horizontal shelf or ledge within the ultimate pit wall slope. The berm interval, berm slope angle, and berm width are governed by the geotechnical configuration of the slope.



- The overall pit slope angle is the angle at which the wall of an open pit stands, as measured between the horizontal and an imaginary line joining the top bench crest with the bottom bench toe.
- The angle of repose or angle of rest is the maximum slope at which a heap of loose material will stand without sliding.
- Haul road width is governed by the required capacity of the road and type of haulage unit. The grade may be defined as the inclination of the road in terms of degrees from the horizontal or percentage of rise to the horizontal.

Pit layouts

- ▶ In a surface mine, the mine layout mainly refers to the position and progress of the benches and faces over the mine life from the first cut (i.e., box cut) to the final cut, i.e., when the ultimate pit configuration is reached.
- ▶ The layout of a mine depends mainly on
 - ❖ Shape of the ore-body - massive, bedded/tabular, vein, pipe, columnar, etc.
 - ❖ Extent of the ore-body
 - ❖ Dip of the ore-body - horizontal, flat, inclined, steeply inclined, sub-vertical, etc.
 - ❖ Depth of occurrence of the ore-body - outcropping, shallow lying, deep seated, etc.
 - ❖ Type of the deposit - sub-surface, hilly, cap type, etc.
 - ❖ The mining system vis-à-vis the equipment being (or to be) used

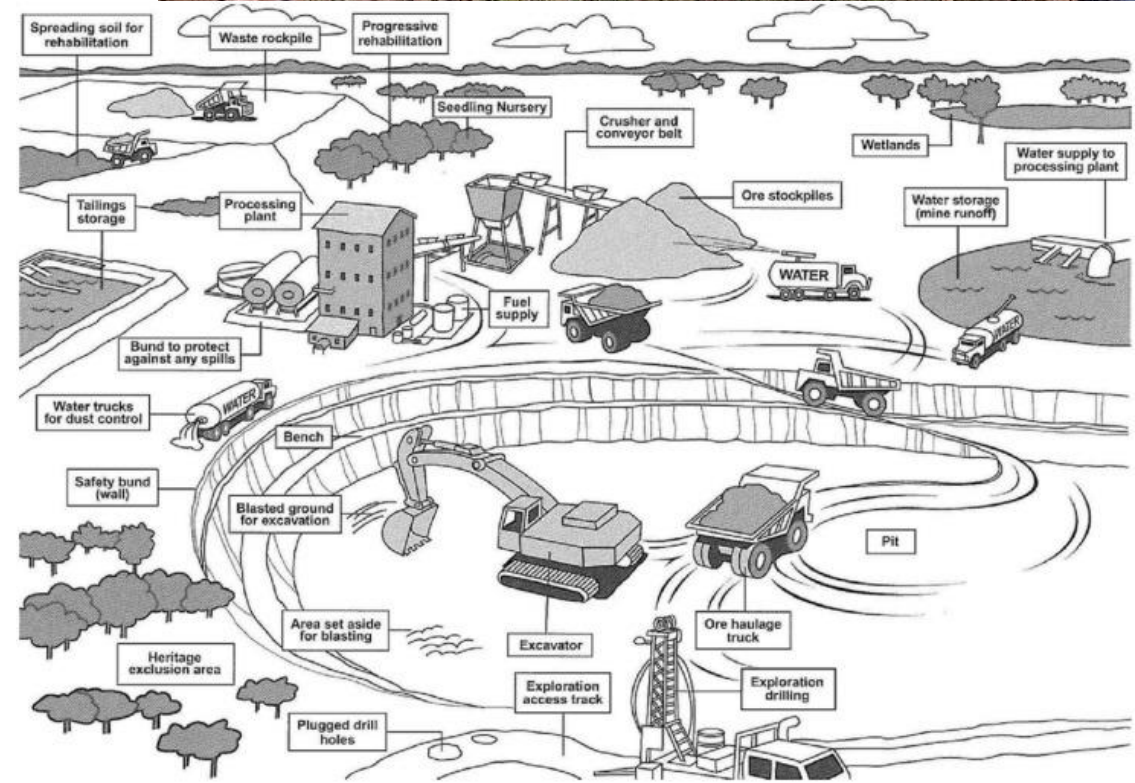


Mine access

- ▶ Access to the deposit is developed to facilitate movement of men and machinery to mineral bed, haul the waste rock and mineral out of the pit and maintained all weather routes to the pit.
- ▶ For the movement of wheel or crawler mounted vehicles, the trenches are to be developed on operating gradient of 1 in 16 for access to the operating benches.

Other considerations are

- ▶ Should be located to minimize the cost of hauling to dump or processing plant
- ▶ Should cross through stable formation; free from geological disturbances
- ▶ Should be located to cater the need of the total block over the life span
- ▶ Should have proper drainage
- ▶ Minimize digging and waste handling



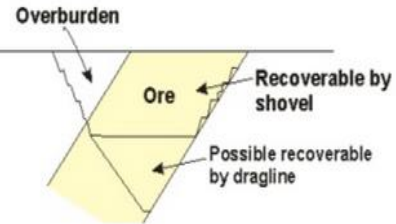
Various open-pit and orebody configurations



Flat lying seam or bed, flat terrain
(Example platinum reefs, coal).



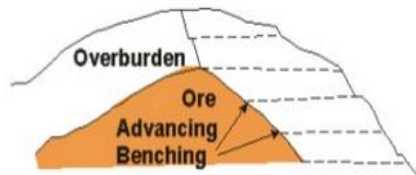
Massive deposit, flat terrain (Example iron-ore or sulphide deposits).



Dipping seam or bed, flat terrain
(Example anthracite).



Massive deposit, high relief
(Example copper sulphide).



Thick bedded deposits, little overburden, flat terrain (Example iron ore, coal).

MINE LAYOUTS

The dependency of mine layout on the shape and extent of ore body is shown in Figure-1.

In case of a deposit that is tabular, inclined and has a large extent in strike direction, faces are laid in a stepped manner oriented in the strike direction. Benches are advanced towards dip direction with approaches at both the ends (Figure-1A).

In the deposits that have limited extent on the strike compared to the dip direction, the layout is restricted by the shape of the deposit (Figure-1B).

If the deposit is a massive one with axial symmetry, the excavation starts at the centre and the benches have concentric circular shape (Figure-1C).

In case of an asymmetric massive deposit, the layout is markedly influenced by the shape of the deposit, and the box-cut starts from the easy mineable location (Figure-1D).

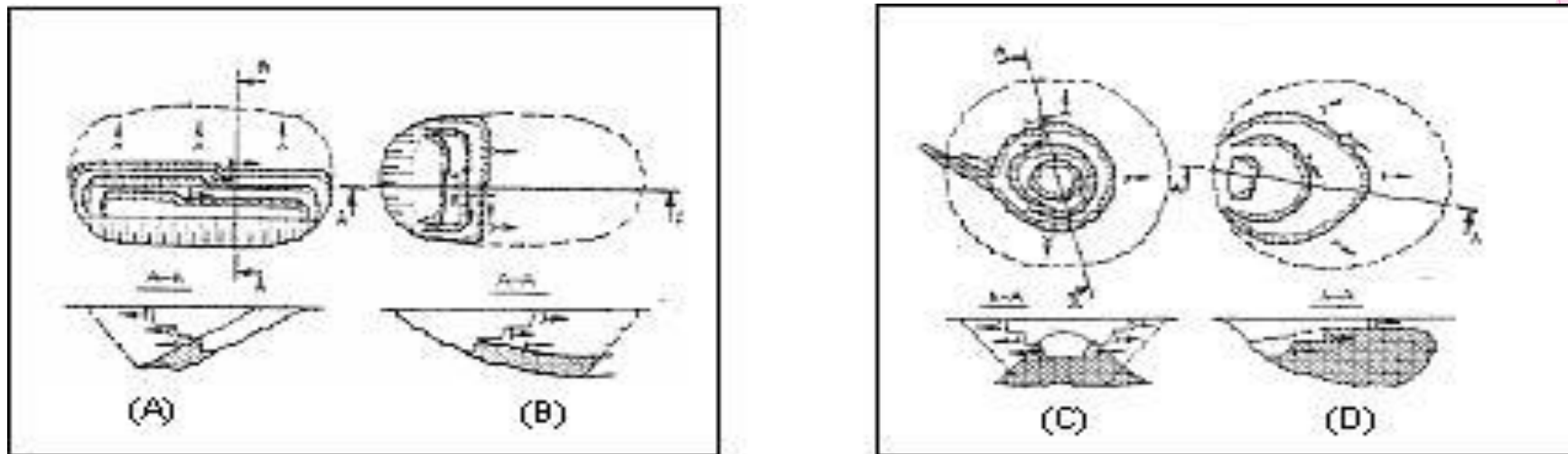


Figure 1 - LAYOUT FOR DIFFERENT TYPES OF DEPOSIT

Layouts for massive deposits with inclination from steep to vertical are shown in Figure-2. In case of **steeply dipping massive deposits**, the excavation is started from the footwall side. The benches advance towards the hangwall side and the approach ramps are placed preferably on the footwall side (Figure-2A). When the **deposit is vertical or nearly vertical**, the deposit is opened up in the centre and the benches move towards the boundary (Figure-2B).

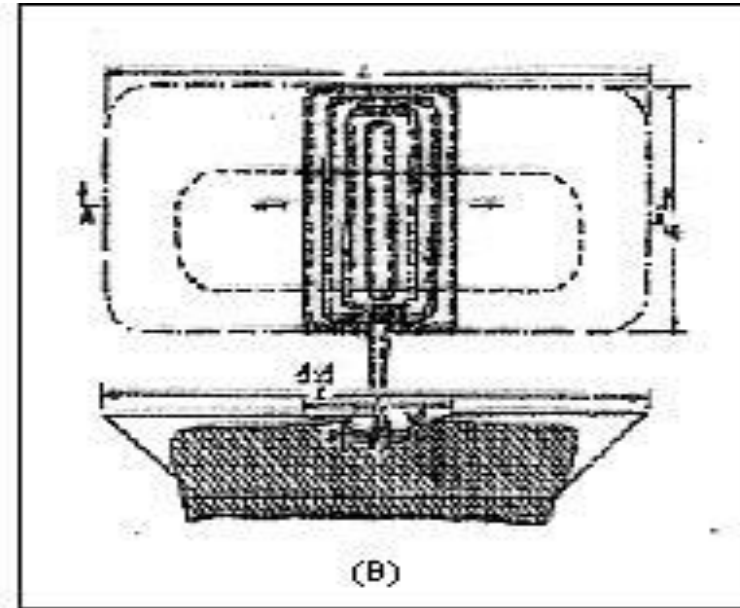
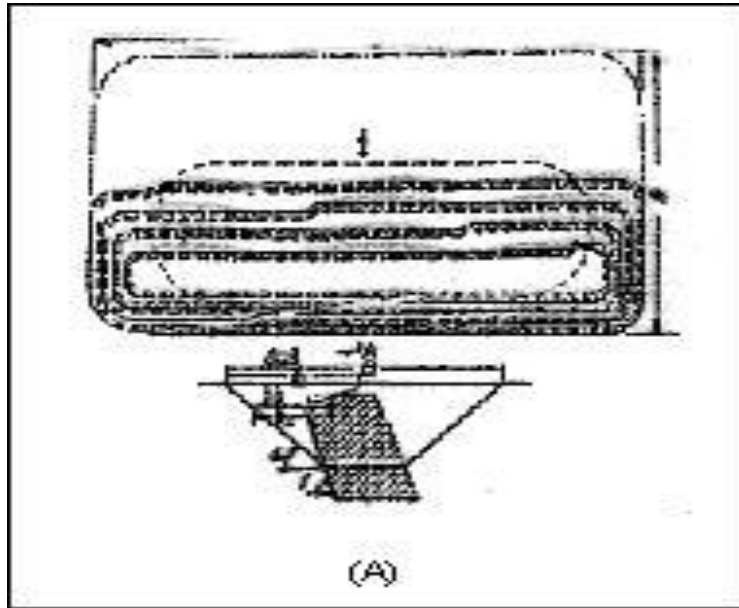


Figure 2 - LAYOUT FOR MASSIVE STEEPLY DIPPING DEPOSIT

In Figure-3, a layout of a mine mining two thick steeply dipping seams is shown. The footwall provides the permanent approach road.

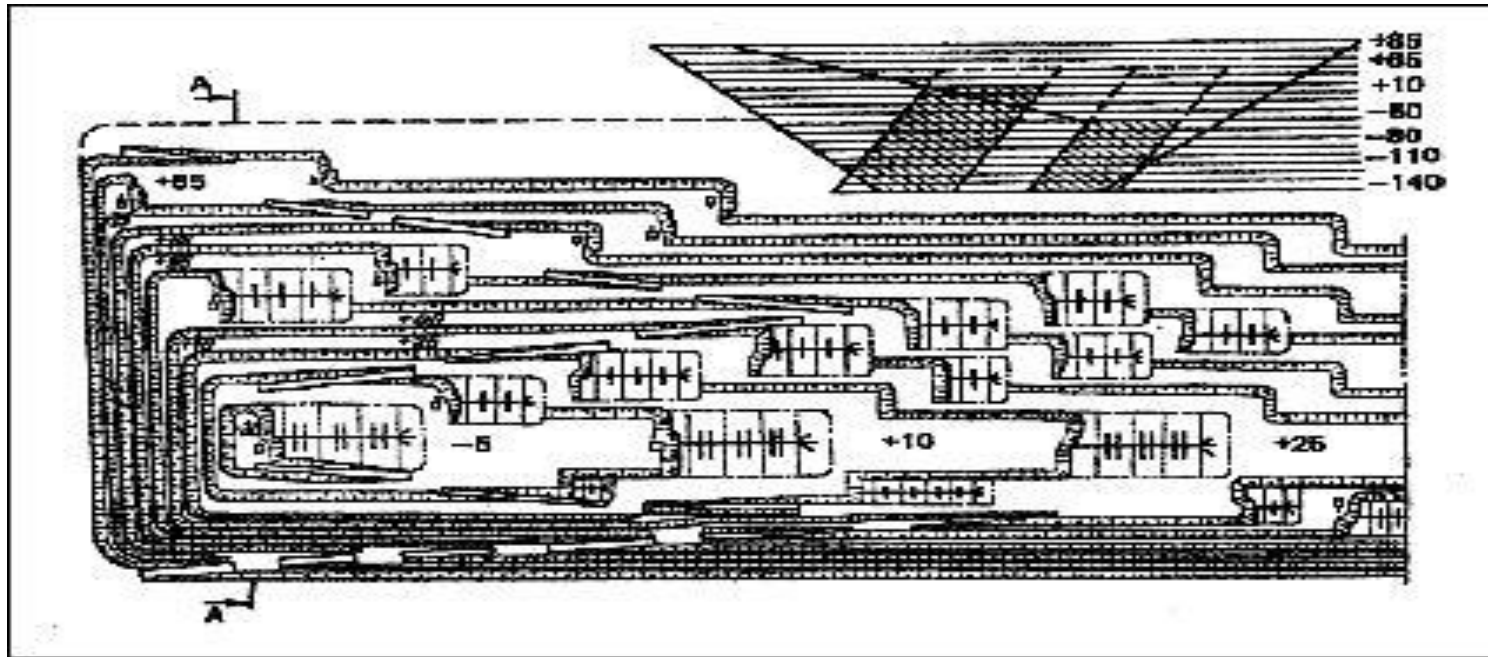


Figure 3 - LAYOUT FOR STEEP AND THICK SEAMS (TWO SEAMS)

In Figure-4 and Figure-5, mine layouts with the haulage system are shown. Layout of a mine in which the development is taking place from rise to dip side, the benches are oriented in the strike direction, and the approach ramps for the upper benches are separate from that of the lower benches is shown in Figure-4A. However, if the deposit is flat or massive, a layout with symmetrical development of benches from the centre with approach road bifurcating from one side of the deposit may be practiced (Figure-4B).

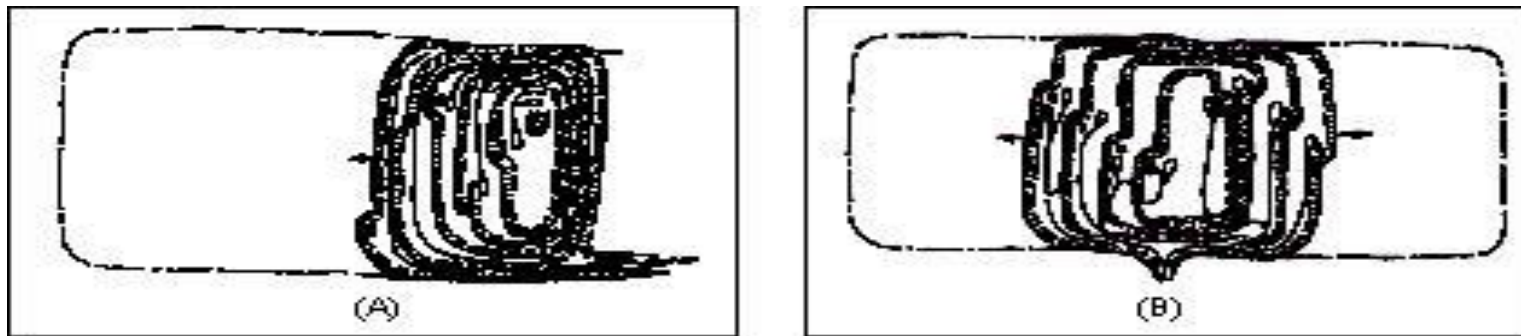


Figure 4 – MINE LAYOUTS WITH HAULAGE RAMP SYSTEMS

Three variants of the same type of layout upto four benches for an inclined bedded deposit are shown in Figure-5.

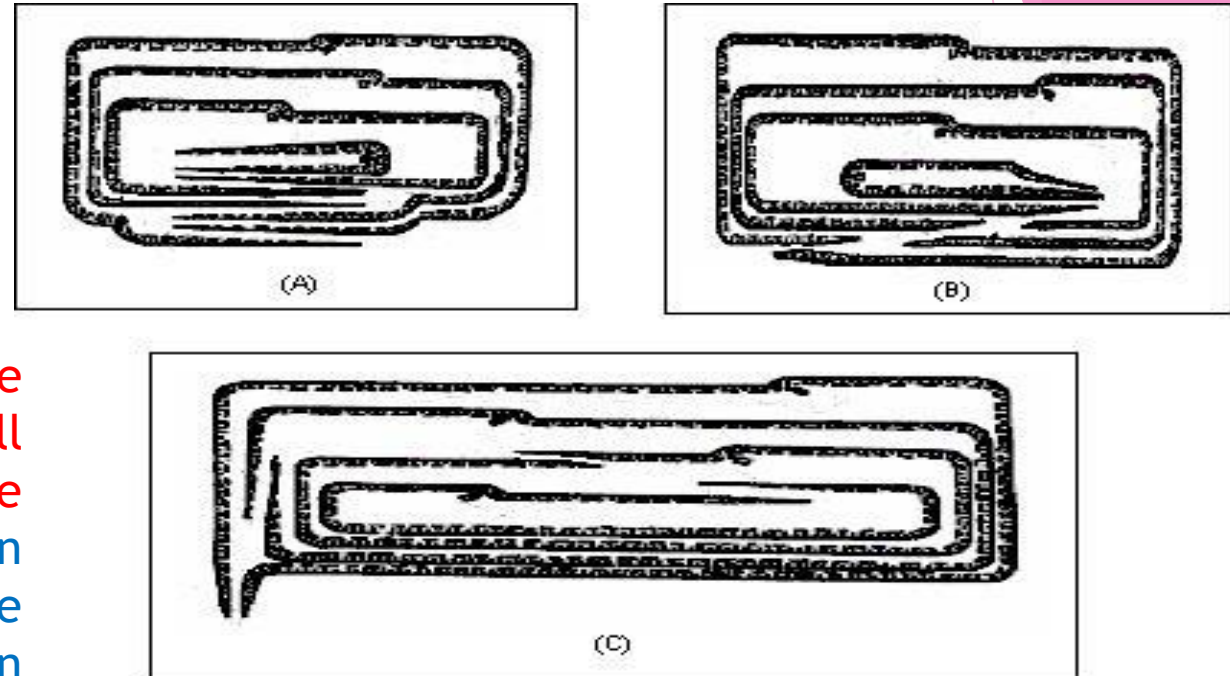


Figure 5 – LAYOUTS FOR INCLINED BEDDED DEPOSITS WITH HAULAGE RAMP SYSTEMS

The first layout (Figure-5A) shows the haulage track/road around the quarry in all the benches. The main ramp makes three turns to reach the bottommost bench. In the second layout (Figure-5B), the arrangements are somewhat different. In this case the straight ramp goes down upto the third bench where it turns around for the fourth one. In case of the third layout (Figure-5C), the ramp enters to the quarry from one side and entry to the subsequent benches is through ramp from the advancing bench side.

Figure-6A shows a layout for a massive deposit. It is similar to those shown in Figure-5. Figure-6B shows a common spiral layout for a massive deposit. In this case, the main ramp goes around the pit in form of a spiral feeding each of the benches (Figure-7). Unidirectional flow of traffic for such a layout is shown in Figure-8.

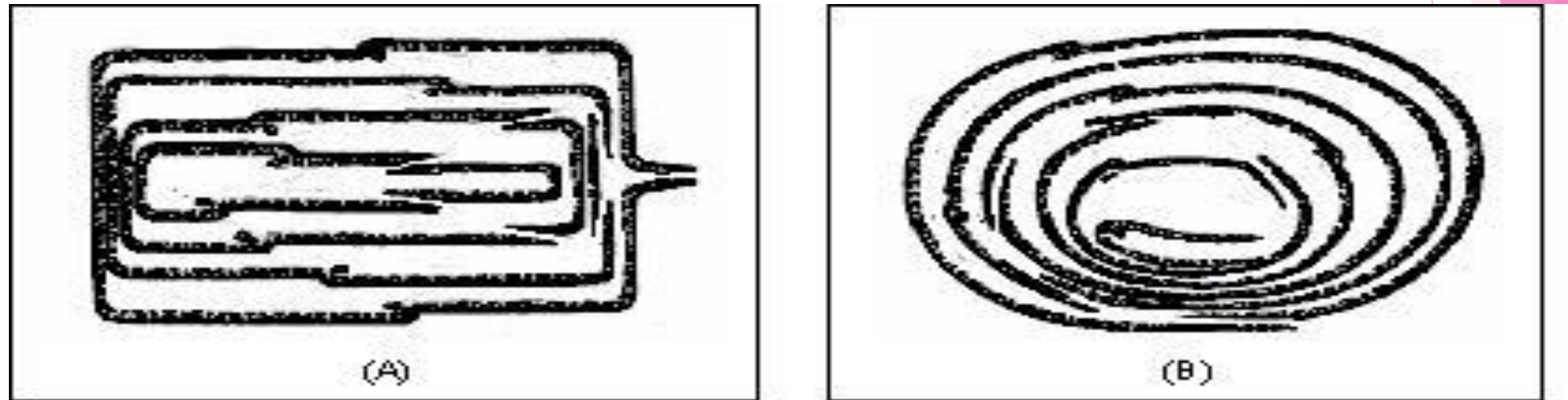


Figure 6 – LAYOUTS FOR MASSIVE DEPOSITS
WITH HAULAGE RAMP SYSTEMS

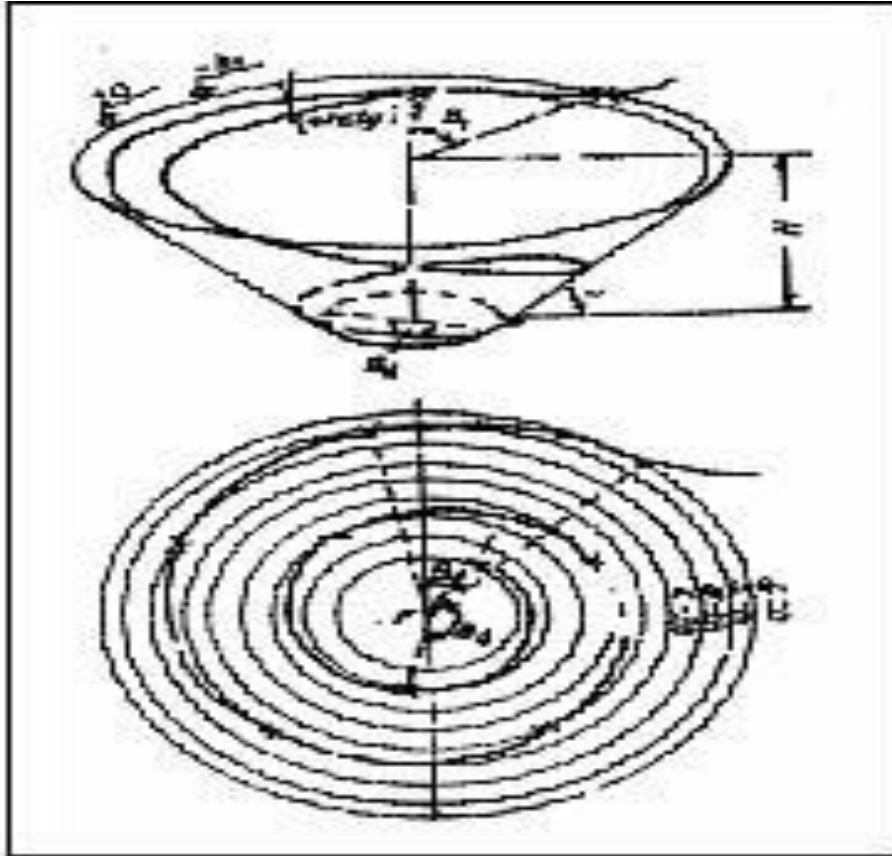


Figure 7 - SPIRAL HAULAGE ROAD

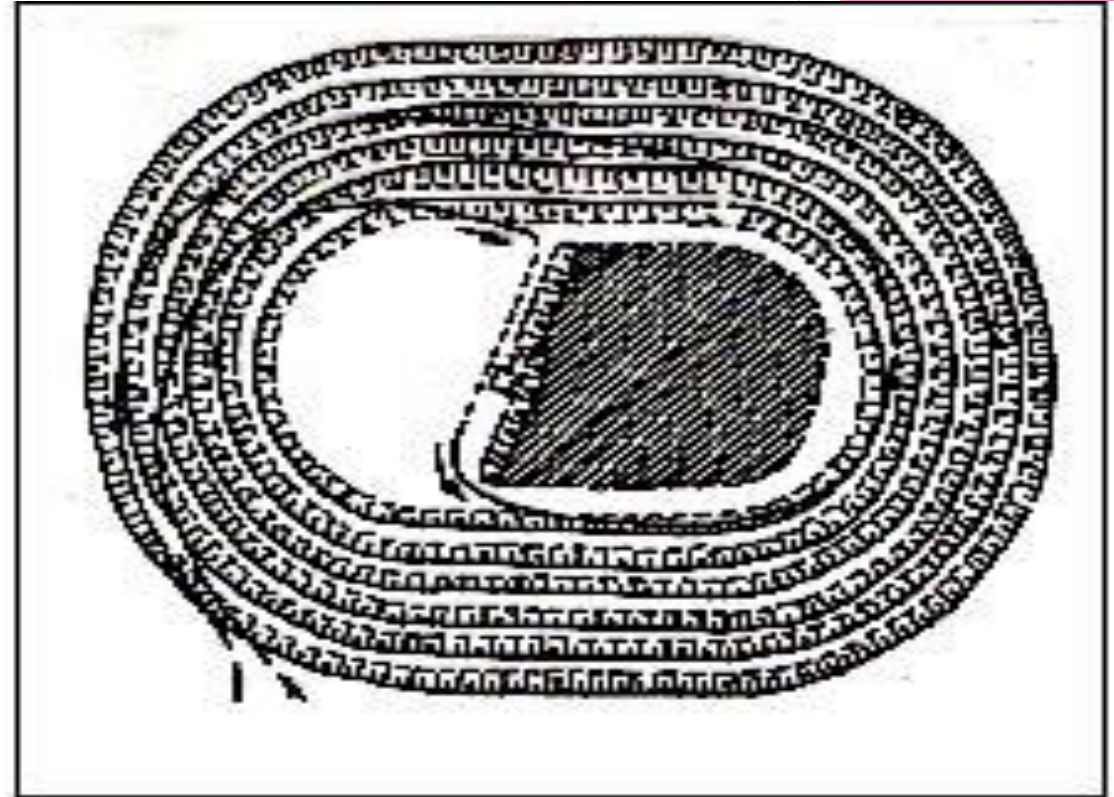
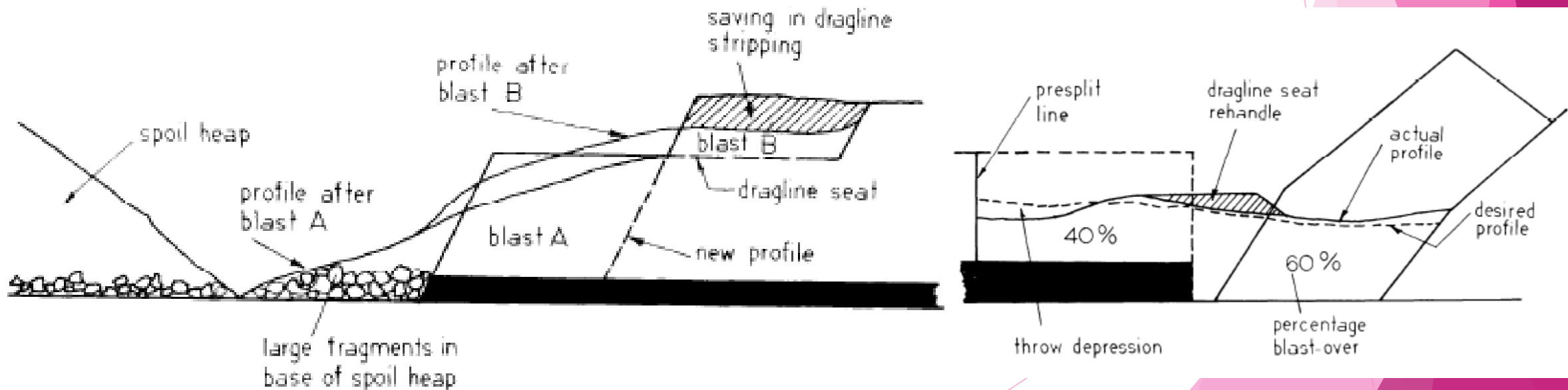


Figure 8 – UNIDIRECTIONAL FLOW OF TRAFFIC WITH A SPIRAL LAYOUT

Layouts with Direct Casting of Overburden

In the following layouts, the overburden is removed by dragline casting and the extraction of mineral is carried out by shovel - dumper combination. A number of layouts highlighting the changes in layout influenced by variations in geometrical shape of the mineralized zone, topography, shape of the deposit and the number of entry have been shown in Figure-9.



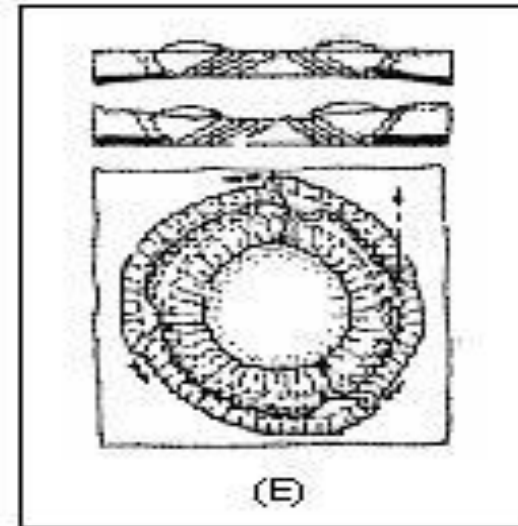
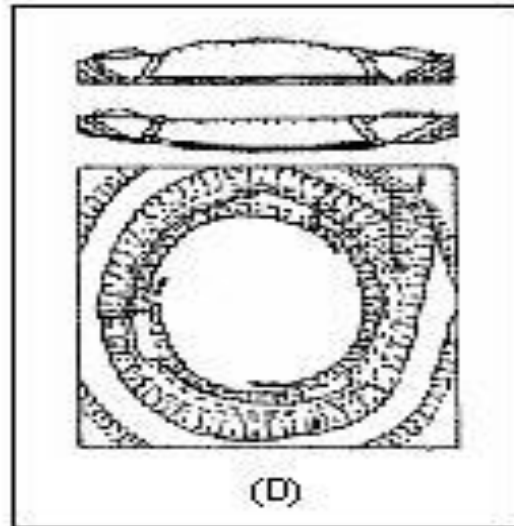
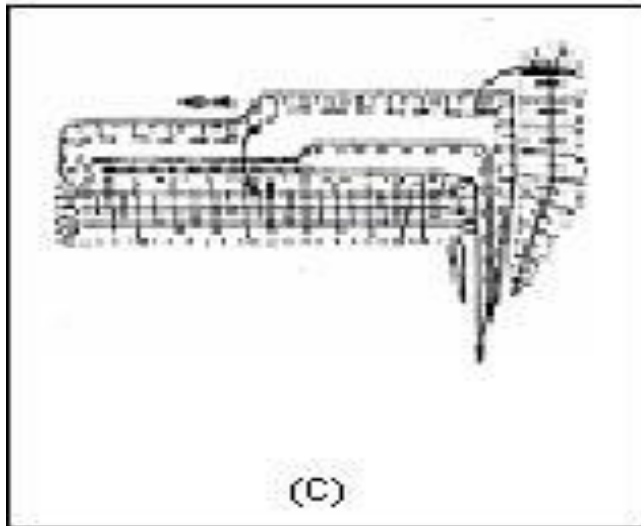
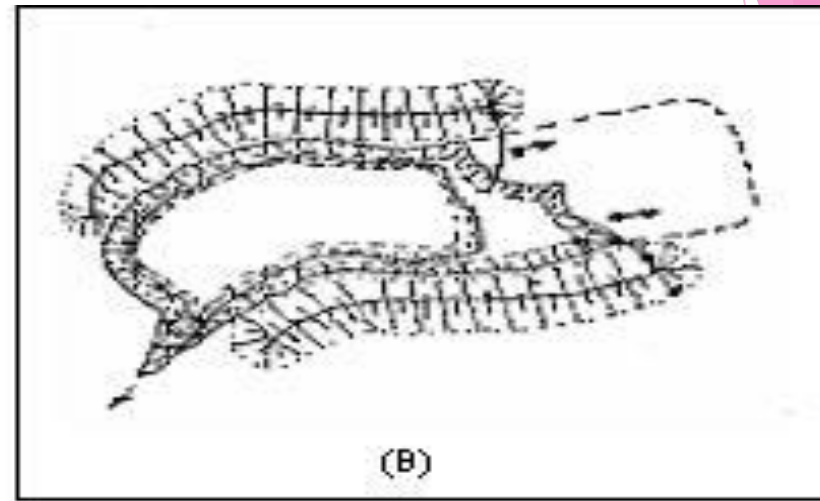
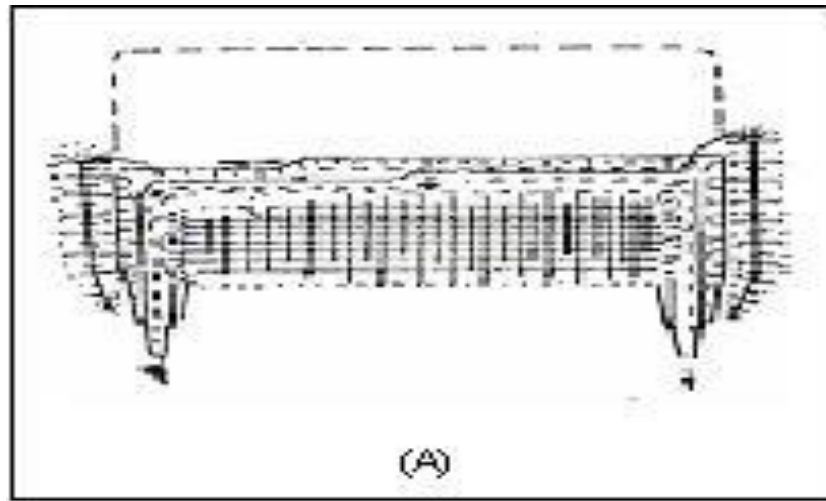


Figure 9 – LAYOUTS WITH STRIPPING BY DIRECT CASTING

In Figure-9A, there are two approach roads for the mineral transport, whereas the overburden is directly cast to the spoil heap. On both sides, the spoil heaps have been created to accommodate the approach roads. Figure-9B shows a rather narrow cut that follows the deposit configuration. Figure-9C shows a layout similar to that shown in Figure-9A but with only one approach ramp. Figure-9D and Figure-9E show the axis symmetric nature of the deposit. In Figure-9D, the thickness of overburden increases towards the centre of the deposit, whereas in Figure-9E, the thickness of overburden decreases towards the centre of the deposit. Such variation of thickness may be due to the topography or the nature of the mineral deposit or due to both.

Two layouts of multiple dragline stripping operation have been shown in Figure-10 and Figure-11. In Figure-10, there are two approach roads at the two ends of the property and the overburden is handled by two draglines following vertical tandem operation. In Figure-11, the pit has been opened with three haul roads - two at the two ends and one in the middle of the pit.

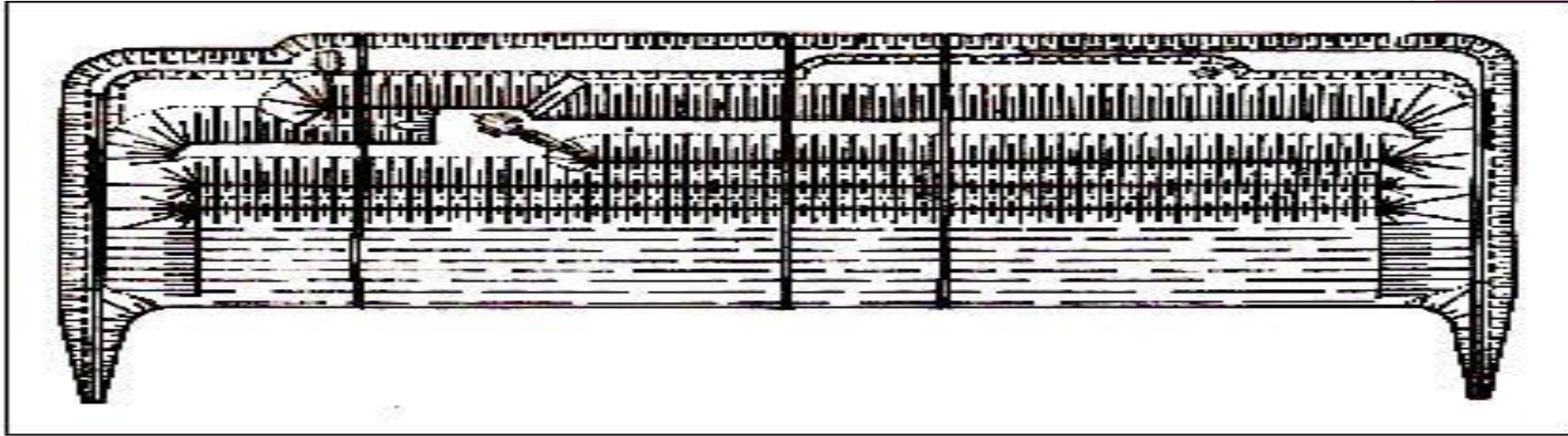


Figure 10 – MULTIPLE DRAGLINE STRIPPING WITH TWO ORE HAUL ROADS

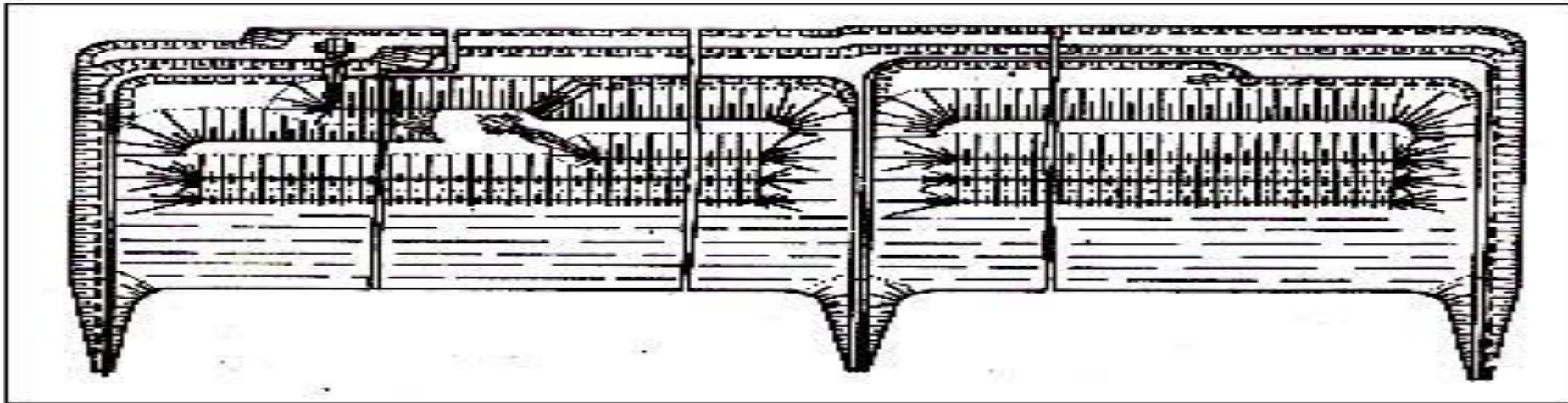


Figure 11 – MULTIPLE DRAGLINE STRIPPING WITH THREE ORE HAUL ROADS

Layouts with Bucket Wheel Excavator Operation

The layouts are basically of two types - overburden transported by around the pit conveyor (APC) system or by cross pit conveyor (CPC) system. In case of fewer benches and normal wind velocity permitting, overburden may be transported across to the spoil heaps/piles by CPC system. In this system, as the transport distance is short, cost of transport is less. However, the transport by APC system is more flexible and can adequately handle difficult depositional features.

Figure-12 shows a number of layouts with transportation of overburden by CPC system. Of particular interest is the change in layouts depending on the shape of the deposit. Figure-13 and Figure-14 show a few layouts of bucket wheel excavator pits where the overburden is transported to the spoil heaps/piles by APC system

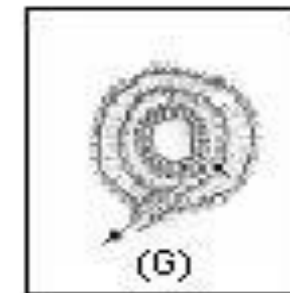
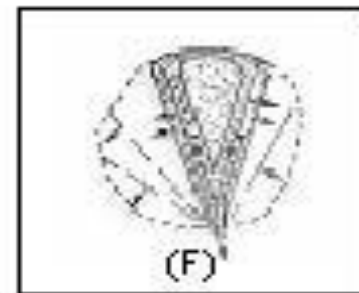
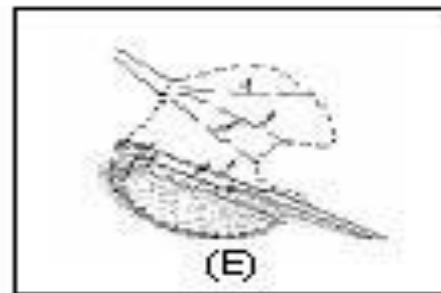
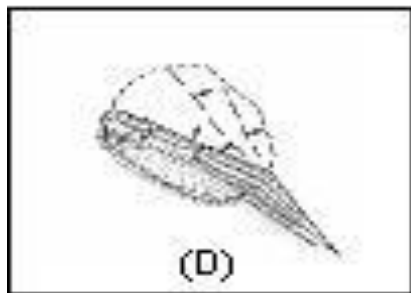
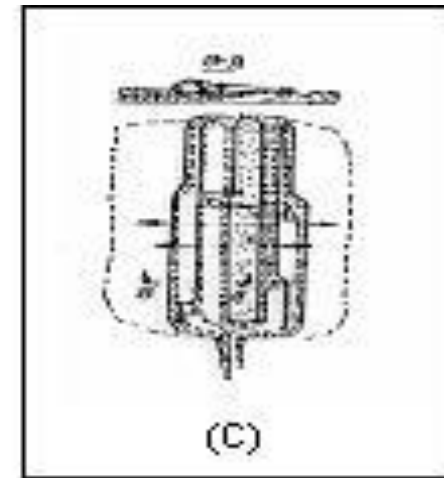
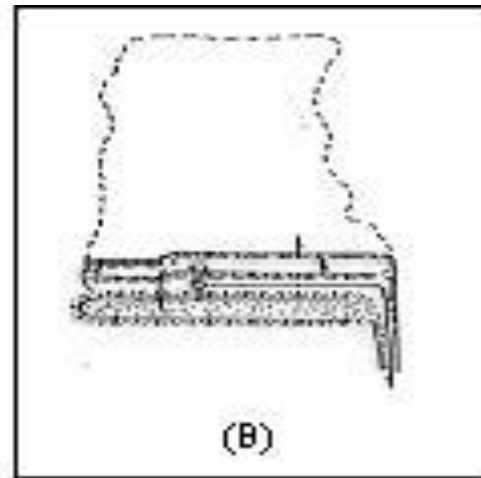
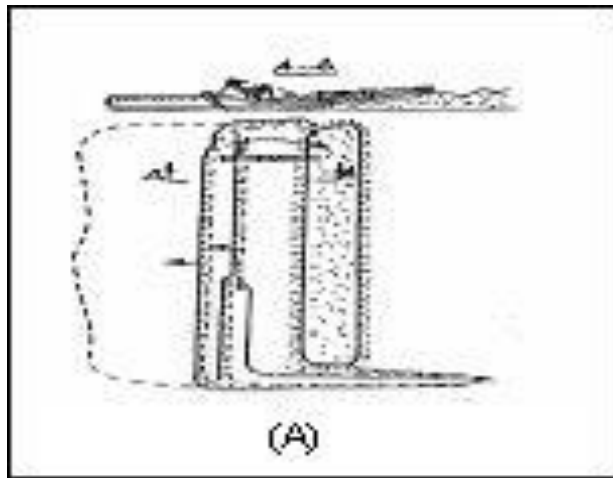


Figure 12 – LAYOUTS WITH STRIPPING BY BWE

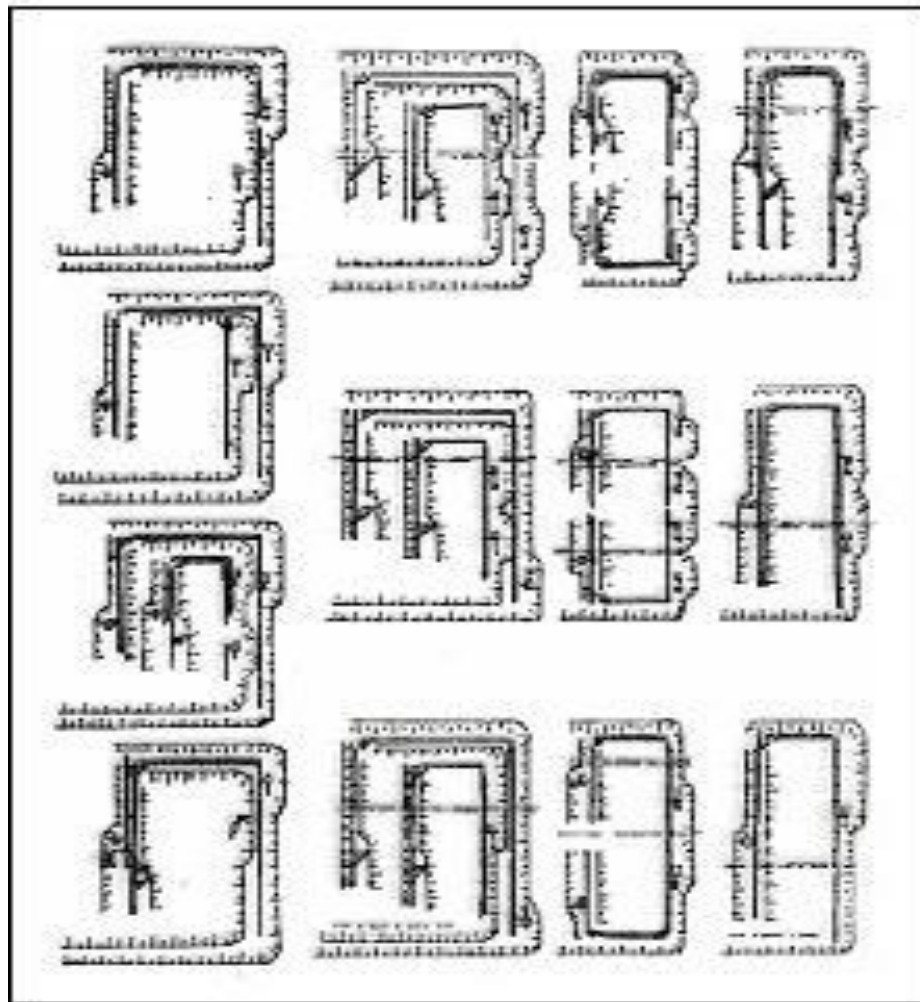


Figure 13 – LAYOUTS WITH BWE OPERATION

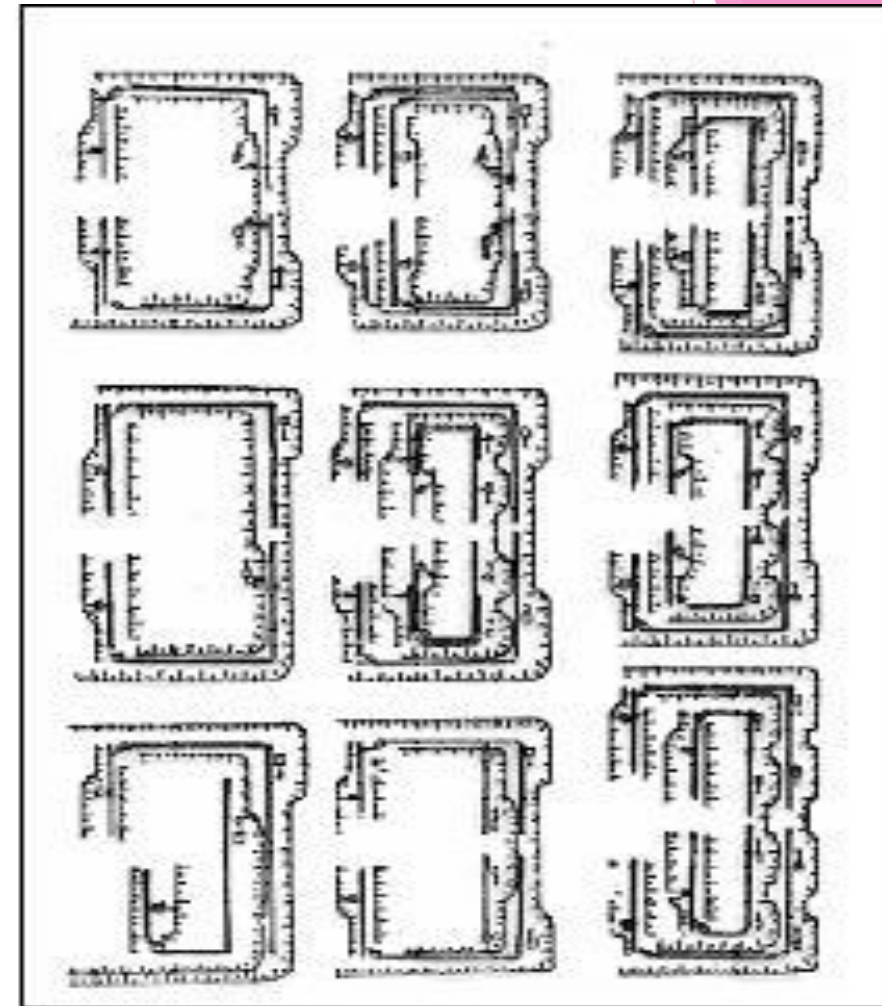


Figure 14 – LAYOUTS WITH BWE OPERATION

Pit expansion

- ▶ The global demand for base metals, combined with diminishing near surface resources, has influenced a number of open pit mining operations to consider the expansion of their mining operations laterally and vertically to recover the deeper reaches of an ore body thereby extending the life of the operation. Today, many surface mining operations are increasingly involving the design and development of deeper and larger open pits.
- ▶ REASONS FOR EXPANSION: *Global Economic Resources on Land*
- ▶ ISSUES AT THE TIME OF EXPANSION OF OPEN PIT MINES
 - ▶ Pit Slope stability
 - ▶ Damage induced by the drilling and blasting operations
 - ▶ Hydrogeological Considerations
 - ▶ Mine Planning for Expansion
 - ▶ **Market and Metal Price Uncertainty**
 - ▶ Haul Roads and Ramps
 - ▶ Waste
 - ▶ Industrial Waste
 - ▶ Mine Environment and Safety
 - ▶ Mine Closure Planning



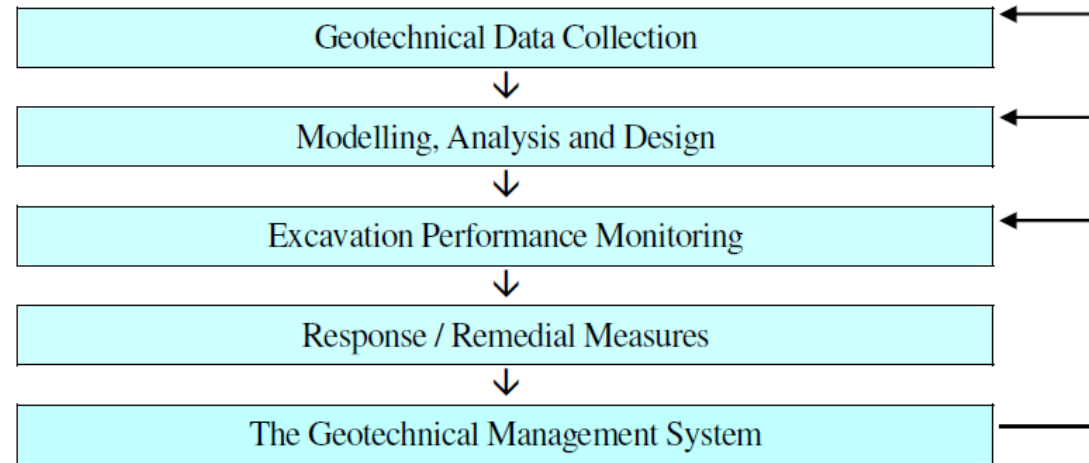
STRATEGIES WHILE EXPANDING THE OPEN PIT MINE

- ▶ Future mining will be shaped in a context where it is necessary to produce at costs that are determined in international competition. The prices of metals and minerals are set by the market but in the long term, there is little doubt that the demand is increasing. Hence, **expansion of existing open pit mine into deep open pit mine will solve the immediate requirement of fulfilling the demand of the market than opening new mine or converting into underground mine.** Therefore, the following strategies can be considered during the expansion of open pit mine to carry out the operations connected with the mining successfully.

- Mining depths increase, and it brings new stability problems. The role of rock mechanics in the design of layouts, cutting sequences, strata stabilization, roof bolting etc. must be a central issue for the future. Production drilling and blasting for controlled fragmentation are two very crucial operations in the ore mining cycle. Improvements in these operations can create significant positive impact on the subsequent operations.
- The environmental requirements affect both energy consumption and management of ore tailings. The discussion of energy consumption is largely linked to global warming and carbon emissions. The cost of emission allowances will be a significant factor to consider, especially for the production located in the more developed countries (Hancock and Sinclair 2008). Improvement of energy efficiency will increase the economic profitability as well as reducing environmental impact.
- The discussion of waste management is about to leave as few footprints in nature as possible. We must not leave toxic substances which leak out into nature and the landscape should be restored as far as possible. The mining companies have to establish a good control of the major water flows in the surrounding rock masses to avoid unwanted leaching and jeopardizing the miners' health and safety. Pollution of mine water is the single most important environmental issue for the mining sector and consequently also effective mine water treatment. New efficient methods and technology needs to be developed as well as relevant guidance for mining in different situations, e.g. climatic conditions, different regulatory and socio economic settings.

- The mines of the future will have a smaller staffing and it is also clear that they will meet a different kind of model for work organization than today. It was predicted that the mining companies gradually will turn to a flat and lean organization with multi-skilled workers who can operate in several areas and functions within the company.
- The detailed guidelines already exist for making technical documents like Geological Report, Mine Plan and Mine Closure Plan, Forest Diversion Proposal and EIA/EMP report. However, guidelines for statutory clearance can also describe the process flow diagram, verifications/scrutiny required and supporting documents for the same at each stage of the process, a model timeframe for processing the application at each stage and the authority accountable for such time bound clearance. These detailed guidelines will make assessment of the application much more objective thereby vastly reducing the time spent in scrutiny and verification. A percentage of the mined out area so reclaimed as forest area may be treated as land made available for compensatory afforestation for future mining projects of the company (Srivastava, 2012). Legislative reforms will be required for transfer of such reclaimed mined out area to forest department. This will galvanize reclamation of mined out area and also OB dumps.

The Geotechnical Management System is dynamic (meaning that it is updated continually) and comprises the following general activities (as shown in Fig.



Under normal conditions the suggested hazard identification process includes:

- **Borehole extensometer data loggers alarmed** with triggers set at levels above the current readings and if triggered will transmit a text message to cell phones of supervisory personnel. Data is downloaded and reviewed on a daily basis.
- **Inclinometer measurements using time displacement plots** and cumulative displacement plots. Results of inclinometer monitoring are communicated by email on a weekly basis as soon as the data has been processed.
- **Wall prism monitoring on a weekly basis by total station.** Individual or groups of prisms may be monitored at greater or lesser intervals as may be notified from time to time.
- **Crusher personnel inspect the crusher slot area on a daily basis to visually assess the cracks in the shotcrete lining.** Any changes in existing cracks or new cracking reported to supervisory staff.
- **Geotechnical personnel walk over accessible parts of the pit, surface facilities area** and crusher area on a weekly basis noting cracks, blast damage or other signs of instability.

Other forms of monitoring in use from time to time include levelling, crack monitors, wire line extensometers.

The strategic mine planning is required to be implemented during expansion of open pit mine which will have one objective of maximizing the value to be realized from extracting a mineral resource. This attempt at value maximization is operationalized through mine planners' suggestions, or "decisions," relating to a number of investment and scheduling alternatives, such as equipment sizes and placements, ore definition (cut-off grade determination), and ore and waste removal sequences (Davis and Newman, 2008). The decisions are updated through time, usually annually, as new information becomes available. The possible strategies are:

- In-Pit Crushing and Conveying (IPCC) – over the steep bench slopes (or) truck haulage ramps.
- Trolley - assisted Truck haulage system (it can be employed only for the ramps in such deep mines where the movement of benches are obsolete).
- Feasibility of Zone-by-zone Using of Dump Trucks of Different Capacity – it depicts the allocation and deployment of particular type and size of loading and transporting equipments zone wise in the mine to optimize the operations when expanding the mine laterally and vertically (Rakishev et al, 2011).

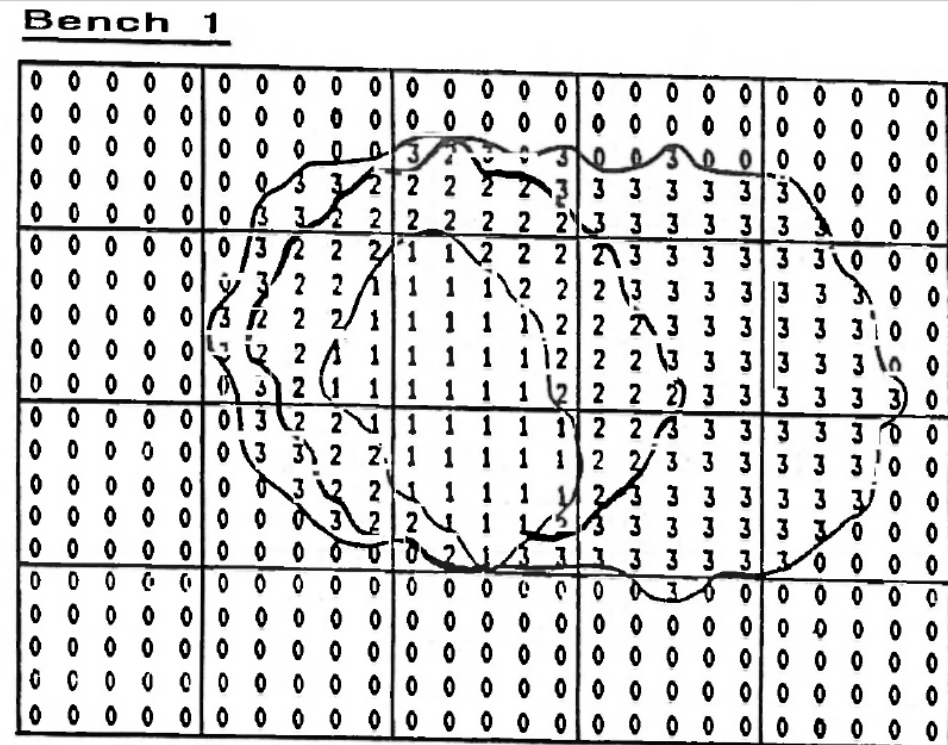
- An open pit mine is developed as a series of nested pits, each larger in area than the previous pit.
- A pushback is the removal of material required to proceed from one pit to the next. The revenue from the ore must pay for the cost of excavating the waste from the pushback and for excavating the ore. The ultimate pit limits should be examined regularly using the latest price-cost forecasts. If a new estimated price-cost relation increases the allowable stripping ratio, the expansion of final pit limit can be considered. But the slope cannot exceed 45° and remain stable so at some point it becomes impossible and/or uneconomic to continue mining. The slopes of a deep pit are cut into a series of locally steep slopes ($> 45^\circ$) each about 5-15 meters high depending on the stability of the rock and the equipment in use. Such steep slopes can become unstable and therefore benches are formed at the bottom of each slope to contain any slope failures. Although parts of the slope are steep, the overall slope angle is low, say $30\text{-}40^\circ$. In some pits the rock may be strong enough to allow “double-benching” where slopes about 20-30 meters high are built. The available equipment must be able to excavate such heights. The objective is to minimize waste excavation, but design and monitoring of such slopes can be difficult.
- The approach to groundwater control can be divided into two general categories; water abstraction in advance of mining (eg. in-pit and out-of-pit production bores); and water abstraction during mining (eg. sumping/trenching into the pit floor and sub-horizontal drainage holes as shown in Fig. 4 and 5). Each method can be used individually, or in combination to produce the required result. Selection of the most appropriate depressurisation method will depend largely on the local and regional hydrogeological conditions, the relative importance of depressurisation to the mine design, and the required rate of mining.

- There is a need to have standardised drilling and blasting patterns that have been determined using well founded and recognised blast design procedures, and that are appropriate to the ground conditions at the mine site. When developing standardized drilling and blasting practices, the mine operator must take into account all factors that control the level of slope damage caused by blasting. Scaling of the batter crest and face following the excavation is an important component of the implementation of the design. Scaling is intended to remove loose blocks and slabs that may form rockfalls or small failures. It is considered good practice to qualitatively and quantitatively monitor the extent of blast damage and evaluate the success of blasting methods as the open pit expands and deepens. Blast monitoring tools include: visual observations, vibration records, noise records, video footage, displacement markers, and complaint records. Blast monitoring results should then be used as part of an ongoing critical review of drilling and blasting to ensure that the blast design is performing to the standards required and is producing the required product.
- In case of existing waste dumps, if the shifting of those waste dumps is inevitable for accommodating the expansion of open pit mine, the possibility of carrying out backfilling with the entire dump material is ascertained. When it is not possible for accommodating complete dumps, then the availability of new area with keeping short lead distance in the mind should be identified.
- The infrastructure facility to carry out the on-field services to the equipments deployed in deeper and isolated parts of the pit should be installed as it would be difficult to bring the equipments always for repairing to the surface (Wolpers, 2013).
- Risk assessment and management should be a part of expansion programme of open pit mine for all the key operations involved in the mining. A team of local people, employees of the mine and representatives from the regulatory agencies (DGMS) is formed to carry out the above exercise before carrying out the expansion of the mine.

Push back operation

- The first step in the practical planning process is to break the overall pit reserve into more manageable planning units. These units are commonly called sequences, expansions, phases, working pits, slices or push backs. In the beginning quite coarse divisions covering periods of several years may be used. They are a preliminary attempt to relate the geometry of mining to the geometry of the ore distribution.

Phase planning should commence with mining that portion of the orebody which will yield the maximum cash flow. Succeeding phases are ordered with respect to their cash flow contribution. Eventually the ultimate pit limits are reached.



This series of steps in manual push back design has been provided by Crawford (1989a).

1. Start with the ultimate pit limit design.

- Develop detailed data of ore grade and stripping distributions for various cutoff grades in zones around the designed pit circumference and in pit shell progression between the beginning surface topography (or pit surface) and the design pit limit. These data should include locations of ore zones (these vary with cutoff) and the impact of the differences between operating and ultimate pit slopes. Of particular interest should be locating high ore grade and low stripping zones on level plan maps and cross-sections.

- Manual planning methods are essentially trial and error approaches.

2. Planning goals typically comprise one or more of the following:

- Maximize NPV economics.
- Provide stable cash flow patterns.
- Uniform ore grade, grade decline curve, or high grading. Frequently high grading is used during the initial investment pay back period. The level(s) of ore grade will be related to the cutoff criteria.
- Uniform stripping ratio or classic stripping ratio curve.
- Uniform total tonnage rate (ore + waste).
- Uniform ore tonnage rate.
- Uniform or variable rate of product output.

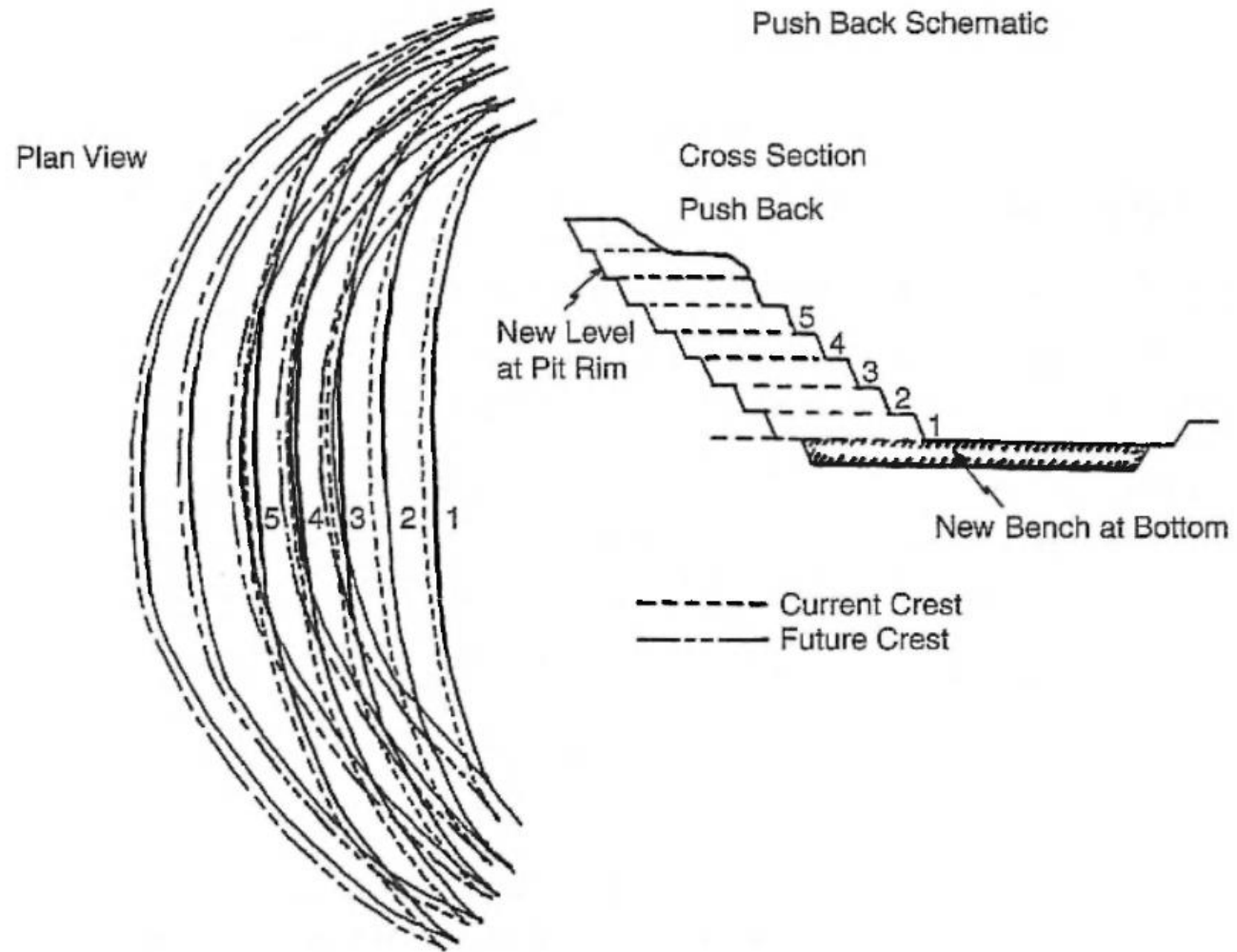
3. Operating design criteria for push back design:

- Operating and remnant bench widths.
- Slopes between operating benches and roads.
- Overall operating slope.
- Road widths and grades.
- Bench height.

Some typical values are as follows:

- Remnant bench width equal to bench height.
- Push back widths normally 200–500 ft depending on size of pit and orebody characteristics.
- Minimum push back widths (single cut passes) about 80 ft for small equipment and 135–150 ft for 25–30 yd shovels and 150–200 ton trucks.
- Road widths 50–80 ft depending on width of equipment. Maximum road grade 8–12 percent.

4. Laying out of one to several push backs. Evaluating whether they satisfy the goals, individually and collectively, is a more or less cut and try process. Normally a push back is laid out according to the operating criteria in plan and cross-section views. The plan view approach is used here. The operating criteria are expressed in geometric parameters (feet and degrees). A useful tool is to make a scaled bench crest and toe pattern including operating bench and road widths.



Push back schematic in plan and section views (Crawford, 1989a).

Thanks